Ogbonna Arinze

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SUMMARY

Senior Fullstack Engineer with 8 years across **Node.js** and the browser, with deep **JavaScript** + **TypeScript** fundamentals and a track record of measurable performance wins (30-40% bundle reduction, 35% load-time improvement, 60-80% lift on settlement and reconciliation throughput). Architect of a 6-app **Nx monorepo** at OurNigeria with sub-second SSE streaming over 708K+ vector embeddings, and a hybrid RAG pipeline tuned for latency. Comfortable in browser and Node runtimes, equally happy hand-tuning a query plan in PostgreSQL, profiling React renders, or shaving startup memory off a Docker Compose stack. Builds ambitious side projects, ships them to production.

CORE COMPETENCIES

- **Node.js Depth:** NestJS, Express-style services, REST APIs, JWT auth, RBAC, SSE streaming, webhook integrations, NodeJS RESTful API design from greenfield.
- **JS/TS Fundamentals:** TypeScript (advanced generics, utility types), JavaScript (ESNext), strong typing, async patterns, runtime memory considerations.
- **Performance Obsession:** 30-40% bundle reduction (rollup-plugin-visualizer, webpack-bundle-analyzer), 35% load-time gain, 40% runtime cache rework, React Profiler, Lighthouse, Core Web Vitals, code splitting.
- **Database and Data:** PostgreSQL with pgvector (708K+ embeddings, hybrid RAG with BM25 + RRF + Cohere rerank), MongoDB schema design, Prisma v7, custom migration tooling.
- **Real-time:** SSE streaming with keepalive heartbeats every 15s, native EventSource API for streamed updates over socket.io at ActivEdge.
- **DevOps and Infra:** Docker / Docker Compose with multi-stage Dockerfiles, health checks, memory limits (API 1GB, Ingest 4GB, Postgres 2GB), 30s graceful shutdown, GitHub Actions CI/CD, AWS, Heroku, Vercel.
- **Frontend:** React, Next.js, Redux, React Query, Tailwind CSS, Radix UI, Storybook.

WORK EXPERIENCE

Genentech (Roche) — San Francisco, CA (Remote)
Frontend Engineer — *May 2023 - Present*

- Co-developed a full-stack **Next.js** chat widget powered by RAG (OpenAI, LangChain, Vercel AI SDK, Chroma) with streaming responses, embeddable on external sites.
- Spearheaded the redesign of the main clinical trial data product, migrating from a **Vue.js** SPA to a **React** Chrome extension with **TypeScript**.
- Optimized application performance through lazy loading, code splitting, image optimization, memoization, and CDN delivery, profiled bottlenecks with **Lighthouse** and Chrome DevTools.
- Introduced **Redux** for client state and **React Query** for server-state caching and request deduplication, refactored class components to functional, tuned the build pipeline.
- Implemented **JWT-based** auth with HTTP-only cookies, RBAC-driven UI, protected routes, and seamless token refresh.
- Architected a reusable design system in **React** and **Radix UI** with **Tailwind CSS**.
- Integrated **Playwright** E2E into **GitHub Actions** CI/CD with Allure Reports on GitHub Pages.
- Maintained the existing **Vue.js** codebase in parallel with the React rebuild without regression.
- Mentored two junior developers (interns).

Descasio — Lagos, Nigeria (Hybrid)
Senior Frontend Developer — *May 2021 - Mar 2023*

- Profiled render bottlenecks with **React Profiler** and **Lighthouse**, cut bundle size **30-40%** via rollup-plugin-visualizer and webpack-bundle-analyzer, achieving **35% faster** initial load.
- Drove a **40% improvement** in runtime performance by reworking caching strategies across the React application.
- Led the design and implementation of a greenfield **React** + **TypeScript** cloud console interface, achieving a **25% boost in user engagement** and accelerating delivery by **30%**.
- Documented React components with **Storybook** and introduced **Jest** + **Enzyme** testing, leading to a **20% drop** in bug reports.
- Established team engineering standards via **ESLint**, **Prettier**, **Husky**, **Lint-staged**, and **GitHub Actions**.
- Mentored junior engineers on frontend best practices.

ActivEdge Technologies — Lagos, Nigeria
Senior React Developer — *Oct 2019 - Sep 2021*

- Adopted the native **EventSource API** for real-time data streaming in place of socket.io, improving update latency and reducing the dependency surface.

- Built a high-volume bank Settlement Solution in a team of 3, handling thousands to millions of transactions daily and boosting settlement rates by **80%**.
- Developed payment modules (MasterCard, VISA, Express-cash, E-commerce) with **React**, **Redux**, **RxJS** for streamed data, and **Axios**.
- Built a real-time account reconciliation tool, lifting daily reconciliation by **60%**.
- Built dashboards with **react-chartjs**, **Redux Toolkit**, and **Ant Design**.

Founders Lair Beijing, China (Remote)
Full Stack Engineer (Contract) *Oct 2020 - Feb 2021*

- Developed **NodeJS** RESTful APIs with testable, maintainable, well-documented code.
- Translated business requirements into efficient **MongoDB** schemas.
- Engineered a CI/CD pipeline for automated API deployment to Heroku and AWS production servers.
- Mentored two interns through structured frontend and backend project work.

SureTrade Business Solutions Nigeria
Frontend Developer *Mar 2018 - Oct 2019*

- Designed **MongoDB** schemas and implemented server-side logic in **NodeJS**, building and testing REST endpoints for multiple production applications.
- Built **SureEdu**, a school management portal, in a team of 5 across staff, student, and guardian modules.
- Authored use cases, translated them into database models, and wrote feature documentation per application.

PROJECTS

OurNigeria ournigeria.ng
Senior Engineer *Feb 2026 - Present*

AI-powered Nigerian budget transparency platform shipped solo to production. Multi-agent LLM pipeline over 700+ documents and 708K+ vector embeddings, performance-tuned end-to-end.

- Architected the platform as an **Nx monorepo** of 6 apps: **NestJS** API, **NestJS** ingestion pipeline, three **Next.js** apps (citizen web, admin dashboard, marketing landing), and a Remotion video service, orchestrated with **pnpm**.
- Built a hybrid **RAG** pipeline combining **pgvector** similarity search with BM25 full-text search, merged via Reciprocal Rank Fusion (K=60), with Cohere rerank-2 and Voyage AI embeddings (1024-dim, 708K+ chunks across 4 vector indexes).
- Implemented **SSE streaming** for real-time chat with keepalive heartbeats every 15s to prevent reverse-proxy timeouts, token-budgeted context (summary + recent + user profile), conversation summarization, and memory extraction.
- Built the document ingestion pipeline supporting PDF (with OCR via Tesseract), XLSX/XLS, DOCX, JSON, and Markdown extractors, processing 700+ documents (959 files) across 37 Nigerian states with zero ingestion errors.
- Designed **PostgreSQL 16** schema with **Prisma v7** and pgvector, with custom migration tooling that filters Mastra-managed chunk tables out of Prisma drift detection.
- Configured Docker Compose production deployment with multi-stage Dockerfiles, health checks, memory limits (API 1GB, Ingest 4GB, Postgres 2GB), and 30s graceful shutdown; integrated **Langfuse** for LLM observability and **Infisical** for secrets.

EDUCATION

University of Nigeria Nsukka, Enugu
BSc Computer Science - 4.1 CGPA *Graduation Date: Sep 2019*